

Generating Probabilistic Aerodrome Forecasts (TAFs) from Numerical Weather Prediction: A Benchmark-Driven Model Review

Spanish airport network, 2020–2025

Working review

Abstract

We study whether machine learning can produce skilful, well-calibrated aerodrome forecasts (TAFs) from numerical weather prediction (NWP) and recent observations, and beat the official human-issued TAF. Using a strictly causal feature frame over 24 Spanish airports (2.5 M METARs, 0.25 M TAFs; ERA5 reanalysis as a perfect-prognosis NWP proxy), we benchmark a ladder of tabular models and a family of sequence/probabilistic models (GRU, LSTM, Transformer, TCN, and a Temporal Fusion Transformer with quantile outputs) on a frozen 2025 test set. The best model is a logistic *stacked ensemble* of {MLP, gradient boosting, TFT} reaching **BSS** +0.348 / **HSS** 0.513, versus the official TAF’s −0.108, and it is near-perfectly calibrated. Driving PROB/TEMPO group generation with it yields an operational TAF at BSS +0.215. Architecture is *not* the limiting factor: data scaling, model class, and the available feature set all plateau near $\text{BSS} \approx +0.33$, indicating a data ceiling set by the ingested NWP fields and the perfect-prognosis setup.

1 Objective

Predict, from information available at issue time T_0 , the hourly aerodrome state over the TAF validity window *probabilistically*, so the model can be scored fairly against the official TAF and drive automatic PROB/TEMPO change-group generation. The headline event is **IFR-or-worse** (sub-instrument ceiling/visibility) — operationally critical and rare ($\sim 3.6\%$).

2 Data and setup

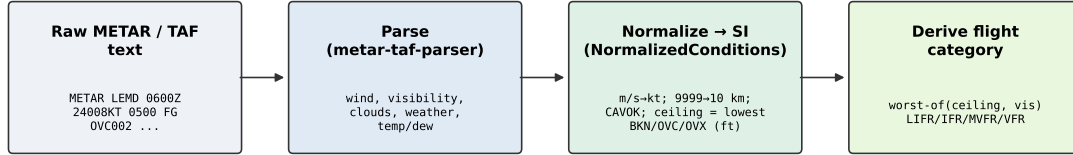
Targets are METAR-derived vis, ceiling, wind and flight category at valid time t . Features are strictly causal (enforced structurally): current and lagged METAR state ($t_0, -1/-3/-6$ h, with tendencies); ERA5 at t and T_0 and the $T_0 \rightarrow t$ tendency (perfect-prognosis NWP); lead time; cyclical hour/day; static airport attributes. Splits are temporal and blocked: train < 2024, validation = 2024, frozen test = 2025; calibration is fit on validation only.

3 Verification: how TAFs and METARs are parsed, scored, and combined

A TAF (a forecast over a window with change groups) and a METAR (a point observation) are different objects, so the verifier reduces both to a common canonical state per hour and scores them with one set of pure functions — the official TAF and any model are judged identically. Four stages: parse $\rightarrow$ normalize $\rightarrow$ expand & align $\rightarrow$ score & combine.

3.1 Parsing and normalization

Raw text is parsed (`metar-taf-parser`) and each report and each TAF trend group is reduced to the same `NormalizedConditions` record: wind (dir/speed/gust), visibility, ceiling, clouds, weather. Normalization coerces units to knots (m/s, km/h); maps ICAO 9999 and `CAVOK` to ≥ 10 km visibility; and computes the ceiling as the lowest BKN/OVC/OVX layer height (or vertical visibility when obscured).



Identical normalization for METAR observations and TAF trend groups → a common canonical state

Figure 1: Parsing/normalization: identical canonical state for observations and TAF groups.

3.2 The key combination: flight category = worst-of(ceiling, visibility)

The flight category collapses ceiling and visibility into one operational state by taking the most restrictive of the two band classifications: LIFR < 500 ft/1600 m, IFR < 1000 ft/4800 m, MVFR < 3000 ft/8000 m, else VFR. This worst-of rule is where cloud/ceiling and visibility *combine*, and it defines the headline event the benchmark optimizes: **IFR-or-worse**.

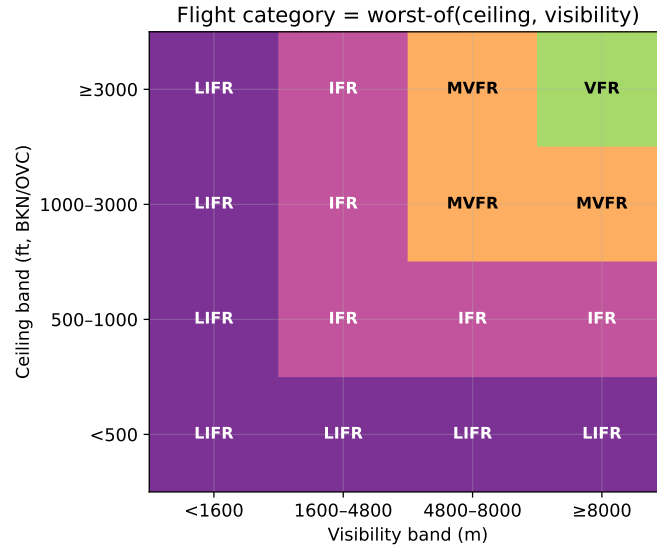


Figure 2: Flight category as the worst-of ceiling and visibility bands.

3.3 Expanding a TAF to an hourly timeline, and aligning observations

A TAF's groups (BASE; instantaneous FM; gradual BECMG; TEMPO/PROB overlays) are walked hour by hour to produce a prevailing state plus any active TEMPO/PROB overlay per hour (`ExpectedHour`); each hour is matched to the nearest METAR within ± 40 min.

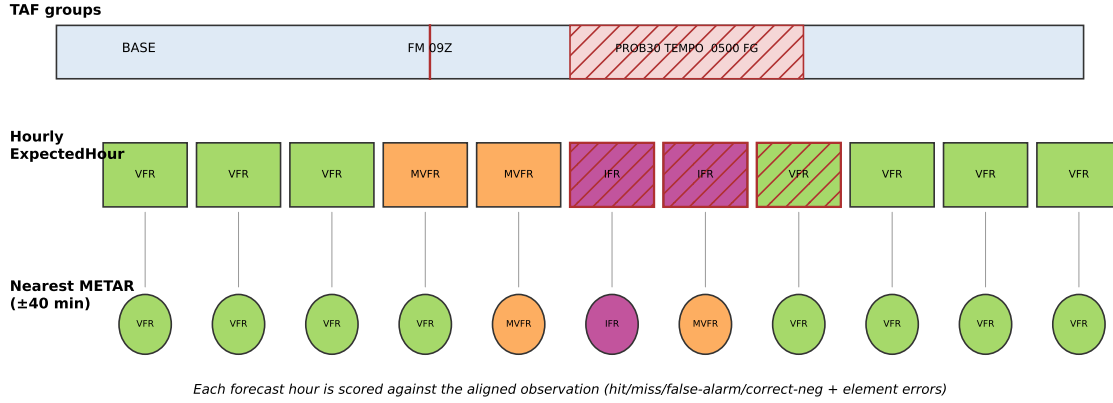


Figure 3: TAF group expansion to an hourly timeline and nearest-observation alignment.

3.4 Per-hour scoring and combination

For each aligned pair the verifier records element errors (wind speed/direction, visibility, ceiling — diagnostic MAEs; temperature is not scored, as a TAF base carries none); a 2×2 categorical contingency for IFR-or-worse (the forecast counts as adverse if *any* of its prevailing/TEMPO/PROB categories is adverse); a hedge-aware probability (`adverse_probability`: prevailing adverse $\rightarrow 1.0$, else PROB30 $\rightarrow 0.3$, PROB40/TEMPO $\rightarrow 0.4$); and a pragmatic 0–3 weighted score. Pooled over the test set these give **HSS** (primary categorical metric and promotion target; plus POD/FAR/CSI), **BSS** (probabilistic, with reliability checking calibration), and a mean weighted score. Thus ceiling $\wedge$ visibility combine into the category, the category defines the adverse event, and HSS/BSS score it; wind and temperature are reported but not optimized.

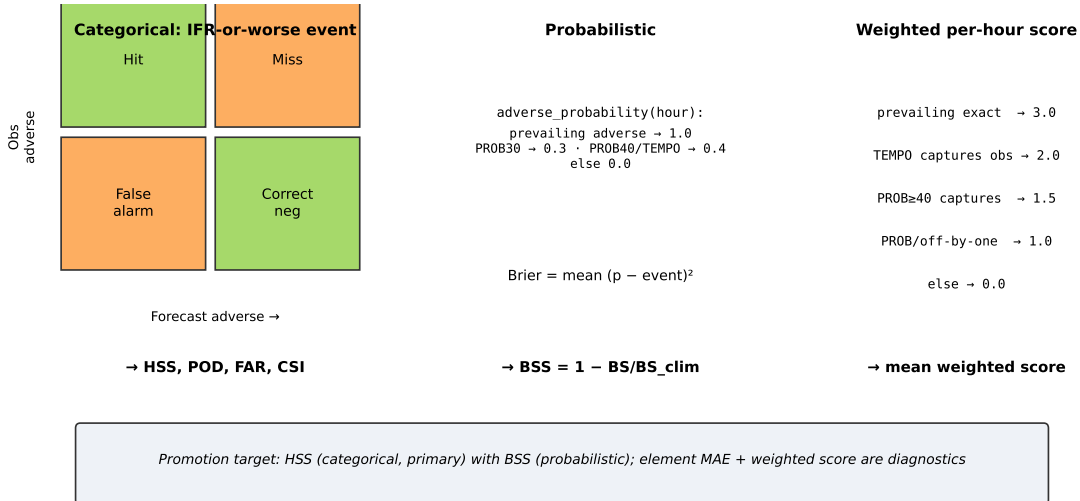


Figure 4: Per-hour scoring combined into HSS, BSS and the weighted score.

A methodological lesson lives here: calibration (isotonic on validation) is decoupled from the decision threshold; a bug where a class-weighted classifier’s raw argmax — not the calibrated decision — drove the forecast category cost the MLP ~ 0.16 HSS until fixed.

4 Models tested

Baselines (persistence, climatology, official TAF); a tabular ladder (linear, gradient boosting, a GPU multi-task MLP); sequence models (GRU, LSTM+attention, Transformer, TCN); a TFT with quantile heads; and ensembles (linear blend, logistic stack). All deep models share one framework, calibration contract and evaluation.

5 Results

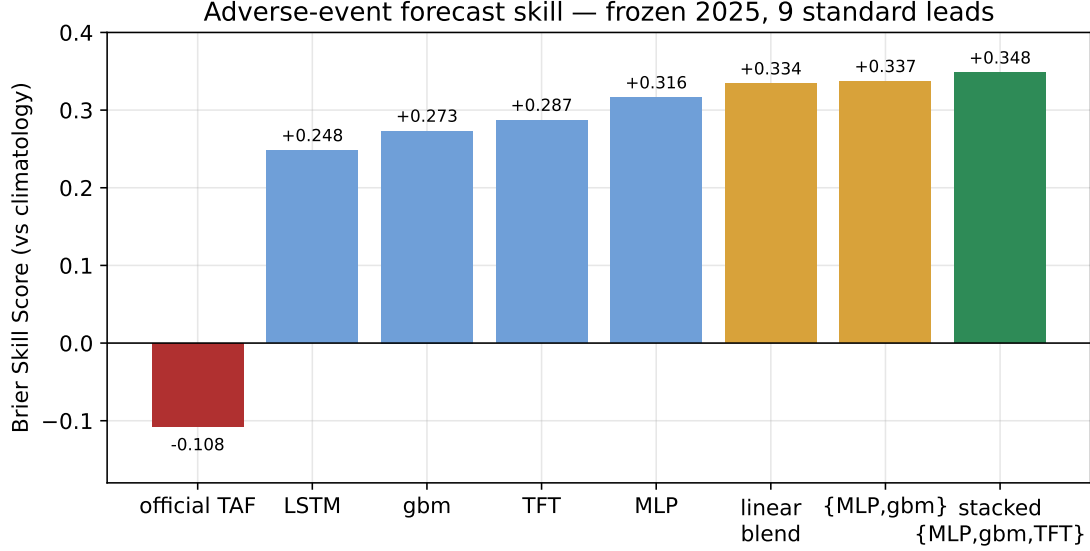


Figure 5: Adverse-event skill on the frozen 2025 test (9 standard leads). Every learned model beats the official TAF (-0.108); ensembling leads.

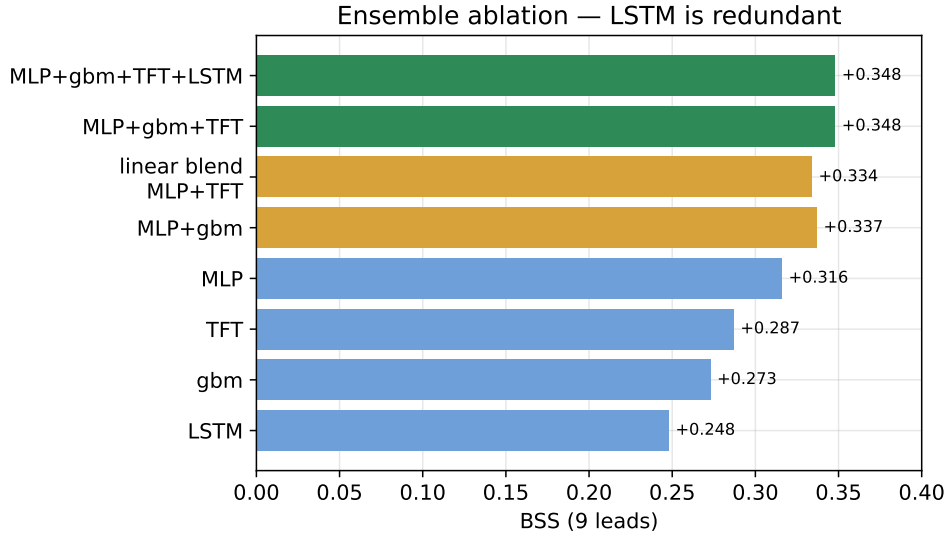


Figure 6: Stack ablation. {MLP, gbm, TFT} matches the full four-model stack (LSTM is redundant); the TFT adds $+0.011$ over the tabular-only {MLP, gbm} stack.

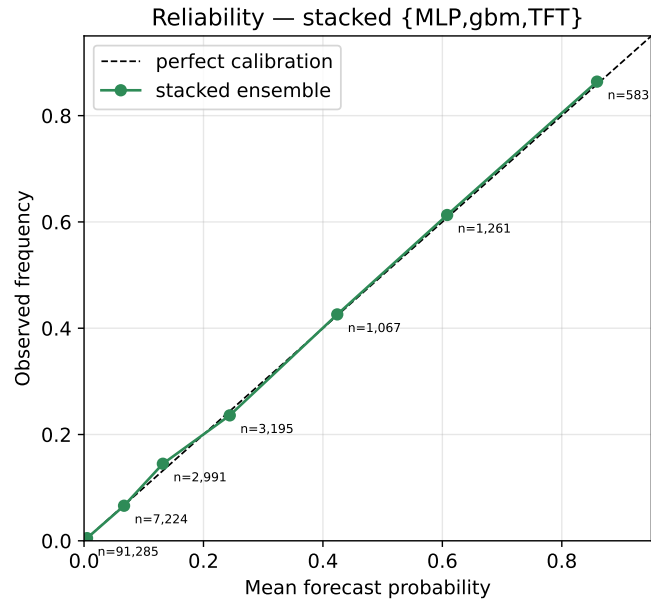


Figure 7: Reliability of the stacked ensemble: forecast probability matches observed frequency to within ± 0.01 .

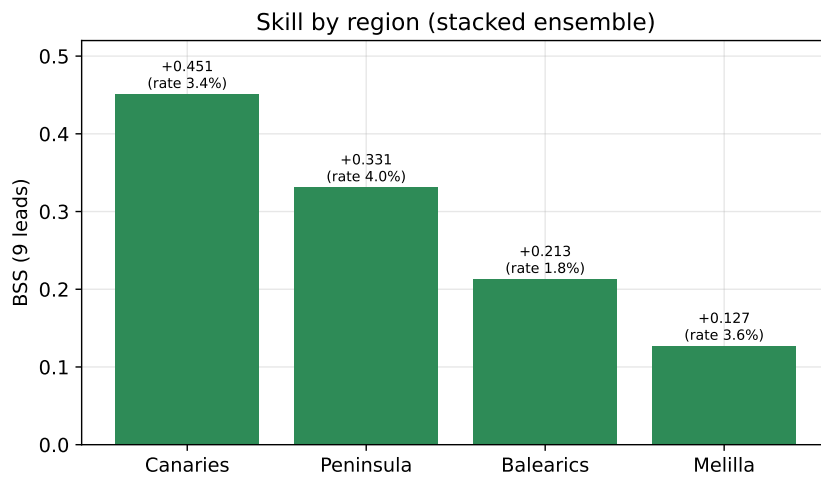
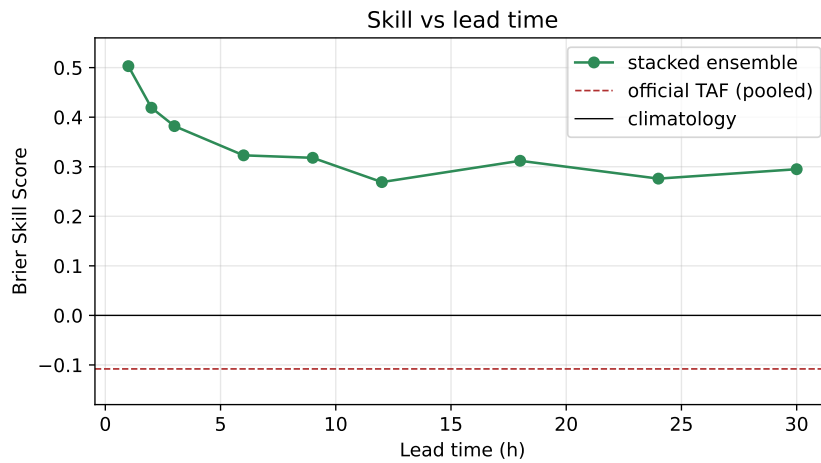


Figure 8: Skill vs lead time (top) and by region (bottom) for the stacked ensemble.

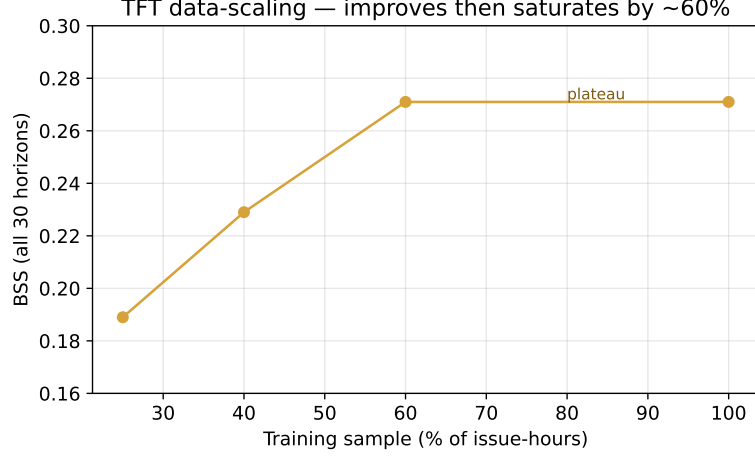


Figure 9: TFT data-scaling: skill improves then saturates by $\sim 60\%$ sample.

6 Generating PROB/TEMPO TAFs

The stacked $P(\text{adverse})$ is quantized to the TAF buckets (none / PROB30 / PROB40 / firm) and combined with the TFT quantile element forecast to emit an hourly timeline with PROB/TEMPO groups. The regenerated operational TAF scores BSS $+0.215$ (vs -0.108). The TFT’s visibility quantiles collapse to “clear” even at q_{10} , so adverse skill lives in the calibrated category probability, not the element distribution; adverse groups use category-representative conditions.

7 Discussion and limitations

Data volume, architecture and feature set all plateau near $\text{BSS} \approx +0.33$ — a data ceiling, not a modelling one. ERA5’s cloud-layer split (the obvious ceiling/fog predictor) was never ingested. The perfect-prognosis setup makes absolute skill optimistic; validation against a genuine IFS/ECMWF reforecast is the credibility milestone. The highest-value next step is richer / genuine-forecast NWP, not a larger network.

8 Conclusion

A calibrated, stacked ensemble of complementary model classes produces aerodrome forecasts substantially more skilful and better calibrated than the official TAF (BSS $+0.348$ vs -0.108) and drives a competitive operational PROB/TEMPO TAF.